

Predicting Band Gaps of SnSe-Based Materials from Composition

Materials Project + matminer + Random Forest

FDP Capstone Project

Machine Learning for Materials and Metallurgical Engineering
IIT (BHU), UGC MM-TTP

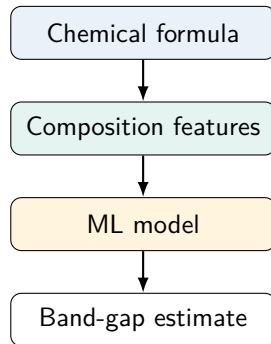
13–18 July 2026

Problem and Motivation

Project question

Can a composition-only model predict whether a Sn–Se-based compound is metal or insulator, and predict the band gap of insulators?

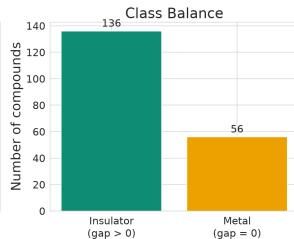
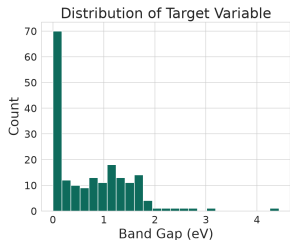
- SnSe is important for thermoelectric and electronic applications.
- Band gap is affected by chemistry, bonding, doping, and structure.
- A fast formula-only model can screen candidates before slower calculations or experiments.



Dataset

Materials Project query

- Compounds containing Sn and Se.
- 220 raw database entries.
- 192 unique formulas after cleaning.
- Target: computed band gap in eV.



Class balance

136 insulators and 56 metals. Accuracy must be compared with this imbalance.

Feature Engineering and Models

Features

- Formula parsed with pymatgen.
- 132 Magpie descriptors from matminer.
- Examples: electronegativity, radius, valence count, periodic column.
- No crystal-structure descriptors used.

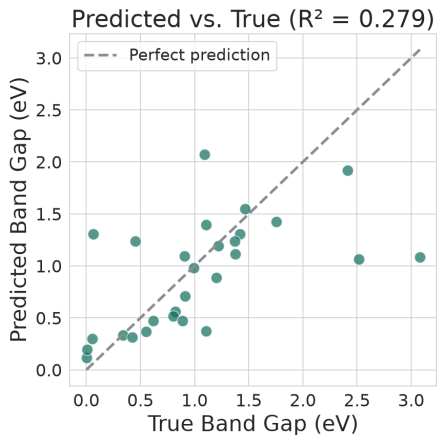
Models

- Logistic Regression classifier.
- Random Forest classifier.
- Random Forest regressor for insulators.
- Primary evaluation: 5-fold cross-validation.

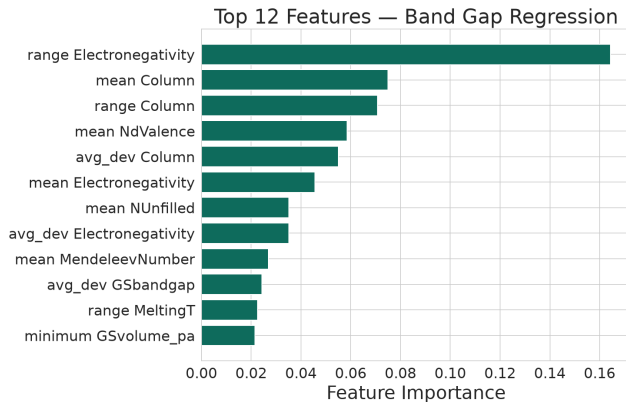
Results

Task	Result
RF classification	$80.2\% \pm 3.7\%$
Logistic classification	$78.1\% \pm 6.5\%$
RF regression MAE	0.357 ± 0.072 eV
RF regression R^2	0.401 ± 0.168

Finding: Random Forest gives the best screening workflow, but regression is not exact.



Feature Importance



Main insight

Electronegativity range is the strongest predictor for the band-gap regression model.

- Large electronegativity contrast changes bonding character.
- Bonding character is related to band-gap opening.
- Feature importance is predictive evidence, not direct proof of causation.

Limitations and Next Steps

Limitations

- Formula-only features cannot distinguish polymorphs.
- 192 compounds is modest for 132 features.
- Class imbalance affects accuracy interpretation.
- Target is computed band gap, not experimental band gap.

Next steps

- Add crystal-structure descriptors.
- Expand to SnS, SnTe, PbSe, PbTe, GeSe, and related compounds.
- Compare more models using repeated cross-validation.

Conclusion

Final message

A composition-only Random Forest model gives a useful first-pass screening signal for Sn–Se-based materials, but it should guide candidate selection rather than replace structure-aware calculations or experiments.

